

EDUCATION

- Ph.D., ELECTRICAL ENGINEERING**2026 —
University of Minnesota - Twin Cities
 - Advisor: Tony Low
 - Focus: Computational Condensed Matter.
- B.S., APPLIED MATHEMATICS, ENGINEERING AND PHYSICS (AMEP)**2022 — 2026
University of Wisconsin - Madison
 - Majors: AMEP + Mathematics + Physics (Honors).
 - GPA: 3.802
 - Focus: QIS, Computational Physics, Numerical Modeling & Optimization

PUBLICATION

- G.-Z. Luo** (2026). A Qubit in an Oscillator: From Ideal GKP Codes to Energy-Scaled Error Mitigation
<http://digital.library.wisc.edu/1793/97588>, Honors thesis in Physics, UW-Madison
- G.-Z. Luo, M. Otten** (2025). Energy-Scaled Zero Noise Extrapolation for Gottesman-Kitaev-Preskill Code
[arXiv:2512.03583 \[quant-ph\]](https://arxiv.org/abs/2512.03583), submitted to npj Quantum Information

GRANTS & AWARDS

- Wisconsin Quantum Institute Research Excellence Award** (\$500)04/2026
 - Best research presentation at the ECE Undergrad Research Symposium
- Outstanding Academic Mentor Award**04/2026
 - For seven semesters of outstanding tutoring through the Center for Academic Excellence.
- AMEP Leadership Prize** (\$1500)04/2026
 - For leadership in the AMEP program, including solar racing strategy, peer mentoring and designing the AMEP shirt.
- Trewartha Senior Honors Thesis Research Grant** (\$1500)11/2025
 - Funded honors thesis research on quantum error mitigation with GKP bosonic codes.
- Jay C. Halls Scholarship** (\$500)05/2025
- UW-Madison Scholarship for Summer Studies** (\$1000)05/2024

TALKS

- Energy-Scaled Zero Noise Extrapolation Extends the Operating Regime of GKP Codes**
Contributed talk, (MAR-A16): Recent Advances in GKP Qubits, APS Global Physics Summit3/16/2026

RESEARCH EXPERIENCE

- Research Assistant**08/2026 —
Low Group, UMN ECE
Minneapolis, USA
- Research Assistant**09/2024 — 05/2026
Otten Group, UW-Madison Physics
Madison, USA
 - Research and honors thesis project on **error mitigation for GKP codes**.

TEACHING

- Academic Mentor (Tutor)**01/2024 — 05/2026
Center of Academic Excellence, UW- Madison
Madison, USA
 - Tutored undergraduate courses in Physics and Mathematics: Pre-Calculus, Calculus, General Physics, Classical Mechanics, Electrodynamics, Linear Algebra, ODE.